

PW EARNERS
DSA in C++

Daily Practice Problems (DPP) — 3
Topic: Loops (For, While, Do-While, Break & Continue)

For Q1 to Q12: Predict the exact output of each program block. If there is a compilation error, infinite loop, or undefined behavior, state the reason clearly.

Q1. (MCQ)

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     for(int i = 1; i <= 3; i++) {
5         cout << i << " ";
6     }
7     return 0;
8 }
```

- A. 1 2 B. 1 2 3 C. 0 1 2 D. 1 2 3 4

Q2. (MCQ)

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int x = 3;
5     while(x--) {
6         cout << x << " ";
7     }
8     return 0;
9 }
```

- A. 3 2 1 B. 2 1 0 C. 3 2 1 0 D. 2 1 0 -1

Q3. (MCQ)

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int count = 0;
5     do {
6         cout << "Loop ";
7         count++;
8     } while (count < 0);
9     return 0;
10 }
```

A. Infinite Loop B. Compilation Error C. Loop D. No Output

Q4. (MCQ)

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     for(int i = 0; i < 5; i++) {
5         if(i == 2) break;
6         cout << i << " ";
7     }
8     return 0;
9 }
```

A. 0 1 2 3 4 B. 0 1 C. 0 1 3 4 D. 1 2

Q5. (MCQ)

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int sum = 0;
5     for(int i = 1; i <= 4; i++) {
6         if(i % 2 == 0) continue;
7         sum += i;
8     }
9     cout << sum;
10    return 0;
11 }
```

A. 10 B. 6 C. 4 D. 3

Q6.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int i = 0;
5     while(i < 3) {
6         cout << i << " ";
7         i += 2;
8     }
9     return 0;
10 }
```

Output:

.....

Q7.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
```

```
4     int i = 1;
5     for( ; i <= 5; ) {
6         cout << i << " ";
7         i = i + 2;
8     }
9     return 0;
10 }
```

Output:

.....

Q8.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     for(int i = 1, j = 10; i < j; i++, j -= 3) {
5         cout << i << " ";
6     }
7     return 0;
8 }
```

Output:

.....

Q9.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int i = 0;
5     while(i = 3) {
6         cout << "PW ";
7         break;
8     }
9     return 0;
10 }
```

Output:

.....

Q10.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     for(int i = 1; i <= 2; i++) {
5         for(int j = 1; j <= 3; j++) {
6             if(j == 2) break;
7             cout << i << " " << j << " ";
8         }
9     }
10    return 0;
11 }
```

Output:

.....

Q11. (Crazy Question MCQ)

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     for(int i = 0; i < 5, false; i++) {
5         cout << "Hello ";
6     }
7     cout << "Done";
8     return 0;
9 }
```

- A. Hello Hello Hello Hello Hello Done
- B. Done
- C. Compilation Error
- D. Infinite Loop

Q12. (Crazy Question MCQ)

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int i = 0;
5     for(;;) {
6         if(i == 5) {
7             break;
8         }
9         cout << i << " ";
10        i += 2;
11    }
12    return 0;
13 }
```

- A. 0 2 4
- B. 0 2 4 6
- C. Compilation Error
- D. Infinite Loop

For Q13 to Q30: Write C++ programs for the following. Ensure proper loop condition checks and variable initializations.

Note: As some questions contain a loop inside a loop, do not use too large of a number while checking your code on your laptop, otherwise it will take a lot of time to execute.

Q13. Take a positive integer N as input and compute the sum of all natural numbers from 1 up to N using a loop.

Sample 1 — Input: 5 $\Rightarrow$ Output: 15

Sample 2 — Input: 10 $\Rightarrow$ Output: 55

Q14. Write a program that takes two integers as input and prints all the even numbers located strictly between them.

Sample 1 — Input: 3 10 $\Rightarrow$ Output: 4 6 8

Sample 2 — Input: 12 21 $\Rightarrow$ Output: 14 16 18 20

Q15. Write a program to take a number as input and compute the product of all its individual digits.

Sample 1 — Input: 413 $\Rightarrow$ Output: 12

Sample 2 — Input: 520 $\Rightarrow$ Output: 0

Q16. Given a positive integer N , write a program to find and print all of its divisors.

Sample 1 — Input: 12 $\Rightarrow$ Output: 1 2 3 4 6 12

Sample 2 — Input: 7 $\Rightarrow$ Output: 1 7

Sample 3 — Input: 15 $\Rightarrow$ Output: 1 3 5 15

Q17. Write a program to find if a given positive integer is a Perfect Number. A perfect number is a positive integer that is equal to the sum of its proper divisors. (A proper divisor of a number is a positive divisor of the number, excluding the number itself.)

Sample 1 — Input: 28 $\Rightarrow$ Output: **Perfect Number**

(*Explanation:* Since $1 + 2 + 4 + 7 + 14 = 28$)

Sample 2 — Input: 12 $\Rightarrow$ Output: **Not a Perfect Number**

(*Explanation:* Since $1 + 2 + 3 + 4 + 6 = 16 \neq 12$)

Q18. Write a program to print all integers from 1 to 500 that are **not** divisible by 3. You must use the `continue` keyword.

Q19. Take an integer N as input and print the Fibonacci series up to N terms. $(0, 1, 1, 2, 3, 5, \dots)$

Sample — Input: 6 $\Rightarrow$ Output: 0 1 1 2 3 5

Q20. Write a program to check if a given number is a Palindrome. A palindrome number reads the same forwards and backwards.

Sample 1 — Input: 121 $\Rightarrow$ Output: Palindrome

Sample 2 — Input: 145 $\Rightarrow$ Output: Not a Palindrome

Q21. Take an integer K as input. Two variables a and b can take any integer value from 1 to K . Write a program to print all possible ordered combinations of (a, b) on a new line.

Sample — Input: 2 $\Rightarrow$ Output:

1, 1

1, 2

2, 1

2, 2

Q22. Write a program that prints all integers from 1 to 500 that contain the digit 2, the digit 7, or both, anywhere within their numerical representation.

Sample — Output includes: 2, 7, 12, 17, 20, 21, 22, ...

Q23. Write a program to iterate through the numbers from 1 to 1000 and print only those numbers whose sum of digits is exactly 17.

Sample — Output includes: 89, 98, 179, 188, ...

Q24. Write a program that takes two positive integers as input and prints all the prime numbers that lie strictly between them.

Sample 1 — Input: 10 20 $\Rightarrow$ Output: 11 13 17 19

Sample 2 — Input: 20 30 $\Rightarrow$ Output: 23 29

Q25. Write a program that prints numbers starting from 1 up to 1000. However, the loop must terminate immediately as soon as it encounters a number whose sum of digits equals 23.

Q26. Take an integer as input and mathematically subtract 293 from it. Compute and print the reversed value of this new resulting integer. Ensure your logic correctly handles negative results.

Sample 1 — Input: 183 $\Rightarrow$ Math: $183 - 293 = -110 \Rightarrow$ Output: -11

Sample 2 — Input: 400 $\Rightarrow$ Math: $400 - 293 = 107 \Rightarrow$ Output: 701

Sample 3 — Input: 293 $\Rightarrow$ Math: $293 - 293 = 0 \Rightarrow$ Output: 0

Q27. Write a program to print all uppercase English alphabets from 'A' to 'Z' along with their corresponding ASCII values using a loop.

Sample — Output includes:

A - 65

B - 66

...

Q28. Write a program that accepts an integer input and finds the strictly next prime number that appears after it. If the user provides an input greater than 500, immediately print "Invalid Input" and terminate.

Sample 1 — Input: 14 $\Rightarrow$ Output: 17

Sample 2 — Input: 499 $\Rightarrow$ Output: 503

Sample 3 — Input: 505 $\Rightarrow$ Output: Invalid Input

Q29. Take a positive integer input from the user and print its prime factors.

Sample — Input: 60 $\Rightarrow$ Output: 2 2 3 5

Q30. Write a program to compute a^b (a raised to the power b) where a is a decimal value and b is an integer (which can be positive, negative, or zero) entered by the user. Do **not** use the `pow()` function from `<cmath>`.

Sample 1 — Input: 2.0 3 $\Rightarrow$ Output: 8

Sample 2 — Input: 2.5 -2 $\Rightarrow$ Output: 0.16

Sample 3 — Input: 5.4 0 $\Rightarrow$ Output: 1

Sample 4 — Input: 4.0 -1 $\Rightarrow$ Output: 0.25

All the best! Keep Coding ♡